COMPETENCY STANDARD

FOR

QUALITY CONTROL MANAGEMENT

(RMG SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standards are developed by a working group comprised of national and international process experts and the participation of experts from the industry to identify the competencies required of an occupation in a particular sector.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (40 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-RMG-QCM-01-G	Carry out workplace interaction	1. Interpret workplace communication and etiquette. 2. Read and understand workplace documents. 3. Participate in workplace meetings and discussions. 4. Practice professional ethics at work.	20
SICIP-RMG-QCM-02-G	Operate in a team environment	1. Identify team goals and work processes. 2. Identify own role and responsibilities within team. 3. Communicate and co-operate with team members. 4. Practice problem solving within the team.	20
		Total Hour	**40**

Sector Specific Competencies (20 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-RMG-SCEM-01-S	Carry out measurements and calculations	1. Plan and prepare. 2. Obtain measurements. 3. Perform calculations.	10
SICIP-RMG-QCM-02-S	Apply green practices in RMG sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
		Total Hour	**20**

Occupation Specific Competencies (300 hrs.)

Code	Unit of Competency	Elements of Competency	Nominal Hours
SICIP-RMG-QCM-01-O	Understand quality management	1. Interpret quality 2. Interpret quality control and quality assurance 3. Identify roles and responsibilities of quality personnel	20
SICIP-RMG-QCM -02-O	Identify garments process operation	1. Identify fabric and garment types 2. Identify sewing machine 3. Identify stitch and seam type 4. Identify types of trim and accessories 5. Interpret garment components	30
SICIP-RMG-QCM-03-O	Identify faults and defects	1. Identify fabric faults 2. Identify trim and accessories faults 3. Identify garment defects 4. Classify garment zones and defects 5. Identify product safety and integrity	30
SICIP-RMG-QCM-04-O	Inspect fabric, trims and accessories	1. Carry out fabric inspection 2. Carry out trim and accessories inspection	20
SICIP-RMG-QCM-05-O	Carry out section wise quality control	1. Perform quality control in sample 2. Perform quality control in washing 3. Perform quality control in cutting 4. Perform quality control in embellishment 5. Perform quality control in sewing 6. Perform quality control in finishing	60
SICIP-RMG-QCM-06-O	Perform final inspection and audit	1. Prepare for final inspection	40

Code	Unit of Competency	Elements of Competency	Nominal Hours
		2. Conduct final product inspection 3. Interpret technical audit system	
SICIP-RMG-QCM-07-O	Interpret TQM and lean management	1. Define TQM And Lean Manufacturing 2. Identify TQM Tools 3. Identify Lean Manufacturing Waste 4. Identify Lean Tools 5. Interpret Six Sigma	50
SICIP-RMG-QCM-08-O	Apply root cause analysis techniques	1. Analyse data and identify root causes 2. Implement root cause analysis techniques 3. Implement corrective and preventive actions 4. Prepare report analysis results	50
		Total Hours	**300 Hrs.**
		TOTAL=	**360 Hrs.**

COMPETENCY STANDARD: QUALITY CONTROL MANAGEMENT

Generic Competencies

UNIT OF COMPETENCY: **CARRY OUT WORKPLACE INTERACTION**	**Nominal Duration:** 20 hrs.	**Unit Code:** SICIP-RMG-QCM-01-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out workplace interaction. It specifically includes task of interpreting workplace communication and etiquette, reading and understanding workplace documents, practicing in workplace meetings and discussions and practicing professional ethics at work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Interpret workplace communication and etiquette	1.1 Workplace codes of conduct are interpreted as per organizational guidelines. 1.2 Appropriate lines of communication are maintained with supervisors and colleagues. 1.3 Workplace interactions are conducted in a **<u>courteous manner</u>** to gather and convey information. 1.4 **<u>Workplace procedures and matters</u>** are comprehended.
2. Read and understand workplace documents	2.1 Workplace documents are interpreted correctly. 2.2 Visual information/symbols/signage are understood correctly and followed. 2.3 Specific and relevant information are accessed from **<u>appropriate sources.</u>** 2.4 Appropriate medium is used to transfer information and ideas.
3. Participate in workplace meetings and discussions	3.1 Team meetings are attended on time. 3.2 Meeting procedures and etiquette are followed. 3.3 Active participation is ensured, opinions are expressed and heard. 3.4 Inputs are provided and interpreted in line with the meeting purpose.
4. Practice professional ethics at work	4.1 Responsibilities as a team member are performed. 4.2 Tasks are performed in accordance with workplace procedures. 4.3 Confidentiality is maintained. 4.4 Inappropriate and conflicting situations are avoided.

Range of variables:

Variables	**Range** (may include but not limited to)
1. Courteous manner	1.1 Effective questioning 1.2 Active listening 1.3 Speaking skills 1.4 Writing skill 1.5 Email etiquette
2. Workplace procedures and matters	2.1 Notes 2.2 Arranging a meeting 2.3 Agenda 2.4 Simple reports such as progress and incident reports 2.5 Job sheets 2.6 Operational manuals 2.7 Brochures and promotional material 2.8 Visual and graphic materials 2.9 Standards 2.10 OHS information 2.11 Signs
3. Appropriate sources	3.1 Human Resources (HR) Department 3.2 Managers 3.3 Supervisors 3.4 Management Information System (MIS)

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace codes of conduct 1.2 Workplace communication and etiquette 1.3 Workplace documents, signs and symbols 1.4 Meeting procedure and etiquette 1.5 Professional ethics 1.6 Responsibilities of team member
2. Underpinning skill	2.1 Demonstrating workplace communication and etiquette 2.2 Interpreting workplace instructions and symbols 2.3 Performing responsibilities of a team member 2.4 Demonstrating active participation in workplace meeting 2.5 Applying professional ethics at work
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace

4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace procedures 4.3 Standard operating procedure 4.4 Workplace documents, signs and symbols 4.5 Projector 4.6 Learning manual

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Interpreted workplace communication and etiquette 1.2 Interpreted workplace instructions and symbols 1.3 Performed active participation in workplace meetings 1.4 Performed responsibilities of a team member
2. Methods of assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **OPERATE IN A TEAM ENVIRONMENT**	**Nominal Duration:** 20 hrs.	**Unit Code:** SICIP-RMG-QCM-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate in a team environment. It specifically includes the task of identifying team goals and work processes, identifying own role and responsibilities within team, communicating and cooperating with team members and practicing problem solving within the team.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1. Roles and objectives of the team are identified and interpreted. 1.2. Roles and responsibilities of team members are identified and interpreted.
2. Identify own role and responsibilities within team	2.1 Personal role and responsibilities are identified within the team environment. 2.2 Reporting relationships are interpreted within team and external to team.
3. Communicate and cooperate with team members	3.1 Other teammates' tasks are identified and support provided when requested. 3.2 The team is encouraged through **sharing information** or expertise, working together to solve problems, and putting team success first. 3.3 Views and opinions of other team members are interpreted and respected.
4. Practice problem solving within the team	4.1 Problems faced at the individual and team level are identified and showed insight into the root causes of the problems. 4.2 A range of solutions and courses of action are identified together with benefits, costs, and risks associated with each. 4.3 The good ideas of others to help develop solutions are recognized and advice sought from those who have solved similar problems. 4.4 It is looked beyond the obvious and not stopped at the first answers.

Range of variables:

Variables	**Range** (may include but not limited to)

1. Sharing information	1.1. Agenda 1.2. Minutes 1.3. progress and incident reports 1.4. Operational manuals 1.5. Visual and graphic materials 1.6. Emails and SMS 1.7. Phone directory 1.8. Policy, procedure and standards 1.9. OHS information

Curricular Content Guide

1. Underpinning knowledge	1.1. Team goals and work processes 1.2. Roles and responsibilities of team members 1.3. Personal role and responsibilities 1.4. Sharing information 1.5. Views and opinions 1.6. Problem solving within the team
2. Underpinning skill	2.1. Identifying team goals and work processes 2.2. Identifying own role and responsibilities within team 2.3. Communicating and cooperating with team members 2.4. Practicing problem solving within the team
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Eagerness to learn 3.5 Tidiness and timeliness 3.6 Environmental concerns 3.7 Respect for rights of peers and seniors at workplace 3.8 Communication with peers and seniors at workplace
4. ‘Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Identified team goals and work processes 1.2 Identified own role and responsibilities within team 1.3 Communicated and cooperated with team members 1.4 Practice problem solving within the team

2. Methods of assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: **CARRY OUT MEASUREMENTS AND CALCULATIONS**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-RMG-QCM-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for carry out measurements and calculations. It specifically includes the task of planning and preparing, obtaining measurements and performing calculations.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Plan and prepare	1.1. Work instructions are confirmed and applied to the job in hand. 1.2. Materials to be measured are identified as per job specification. 1.3. Appropriate **<u>measuring device</u>** is identified and selected based on materials to be measured. 1.4. Specifications are obtained and verified from relevant **<u>documents.</u>**
2. Obtain measurements	2.1 Method of obtaining measurement is selected and applied. 2.2 **<u>Measurements</u>** are obtained using appropriate device in accordance with job requirement. 2.3 Measurements, including area, volume, tolerance and clearance limits, are confirmed and applied
3. Perform calculations	3.1 **<u>Calculations</u>**, using basic operations, for determining material requirement are taken. 3.2 Appropriate **<u>formulas</u>** for calculating quantities are selected. 3.3 Quantities are estimated from the calculation taken. 3.4 Material quantities are calculated, confirmed and recorded within tolerances.

Range of Variables

Variable	Range May include but not limited to:
1. Measuring device	1.1 Measuring tape 1.2 Steel rule 1.3 Calculator 1.4 Sets square
2. Documents	2.1. Technical manuals

	2.2. Specifications 2.3. Sketches 2.4. Drawings 2.5. Charts 2.6. Photographs
3. Measurements	3.1. Length 3.2. Width 3.3. Weight 3.4. Tolerance
4. Calculations	4.1 Addition 4.2 Subtraction 4.3 Multiplication 4.4 Division 4.5 Area 4.6 Volume 4.7 Circumference 4.8 Cubic meter 4.9 Volumetric weight
5. Formulas	5.1 Fractions 5.2 Percentages 5.3 Mixed numbers 5.4 Conversions 5.5 Scales

Curricular Content Guide

1. Underpinning Knowledge	1.1 Measuring devices 1.2 Measurements 1.3 Simple calculation techniques 1.4 Appropriate formulas 1.5 Estimating quantities
2. Underpinning Skills	2.1 Identifying appropriate measuring devices 2.2 Carrying out measurements for appropriate device 2.3 Performing calculations using appropriate formulas
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Communication with peers, sub-ordinates and seniors in workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Map/globe 4.3 Projector 4.4 Stationery 4.5 Learning manual.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified and selected appropriate measuring devices 1.2 Carried out measurements for appropriate device 1.3 Identified and selected correct mathematical formula 1.4 Performed calculations using appropriate formulas
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **APPLY GREEN PRACTICES IN RMG SECTOR**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-RMG-QCM-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply green practices in RMG sector. It specifically includes the tasks of interpreting green concepts, minimizing resources use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 **<u>Principles of green practices in RMG (Ready-Made Garment)</u>** manufacturing are explained. 1.2 Key terms and symbols in RMG production processes are identified. 1.3 **<u>Sources of environmental impacts</u>** during RMG manufacturing activities are identified. 1.4 RMG manufacturing activities use are explained. 1.5 **<u>Ways of mitigating environmental impacts</u>** in RMG manufacturing are explained.
2. Minimize resources use in the workplace	2.1 Water, energy, and raw material consumption in RMG manufacturing are documented. 2.2 **<u>Recyclable and non-recyclable items</u>** in RMG production processes are identified. 2.3 Procedures to reduce resource consumption in RMG manufacturing are implemented. 2.4 Sustainable alternatives in RMG manufacturing are explored and applied.
3. Implement waste management practices	3.1 **<u>Different types of waste in RMG manufacturing</u>** are identified. 3.2 Hazardous waste in RMG production is disposed of according to environmental regulations. 3.3 Green habits to reduce waste in personal and professional life are practiced.

Range of Variables

Variable	**Range** May include but not limited to:
1. Principles of green practices in RMG	1.1 Reducing energy consumption in RMG manufacturing processes. 1.2 6R for waste management in RMG production:

(Ready-Made Garment)	1.2.1 Refuse unnecessary materials and processes. 1.2.2 Reduce resource consumption and waste generation. 1.2.3 Reuse materials within the production process. 1.2.4 Recycle materials for future use in RMG manufacturing. 1.2.5 Recover usable resources from waste. 1.2.6 Repair damaged products or materials instead of discarding them. 1.3 Use of sustainable materials in RMG production with low environmental impact. 1.4 Recycling of materials used in the RMG manufacturing process. 1.5 Sustainable transportation methods for moving goods and materials within the RMG sector.
2. Sources of environmental impacts	2.1 Air Pollution in RMG Manufacturing 2.2 Water Pollution in RMG Manufacturing 2.3 Soil Degradation in RMG Manufacturing 2.4 Noise Pollution in RMG Manufacturing 2.5 Waste Generation in RMG Manufacturing 2.6 Resource Depletion in RMG Manufacturing
3. Ways of mitigating environmental impacts	3.1 Utilizing energy-efficient equipment in RMG production processes. 3.2 Adopting renewable energy sources for RMG manufacturing operations. 3.3 Implementing site protection measures to prevent pollution and reduce environmental. 3.4 Using reusable materials in production processes. 3.5 Choosing sustainable materials with low environmental impact for garment production. 3.6 Using noise-reducing equipment and scheduling production work. 3.7 Optimizing logistics and delivery to reduce transportation emissions and improve overall supply chain efficiency.
4. Recyclable and non-recyclable items	4.1 **Recyclable Items in RMG Manufacturing** 4.1.1 Metal components. 4.1.2 Wooden materials. 4.1.3 Fabric scraps and textile fibers. 4.1.4 Plastic items, including buttons, tags, and packaging materials. 4.2 **Non-Recyclable Items in RMG Manufacturing** 4.2.1 Paints, coatings, and dyes. 4.2.2 Contaminated materials. 4.2.3 Treated wood and other materials.

5. Different types of waste in RMG manufacturing	5.1 Fabric waste from cutting and trimming processes. 5.2 Wood waste from packaging or display materials. 5.3 Metal waste from trims, buttons, zippers, and other hardware. 5.4 Textile waste, including scraps from garment production and defective products. 5.5 Plastic waste, such as packaging materials, labels, and plastic buttons. 5.6 Solid waste from chemical treatments or dyeing processes. 5.7 Packaging waste from raw materials and finished products.

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in RMG (Ready-Made Garment) manufacturing. 1.2 Sources of environmental impacts. 1.3 Ways of mitigating environmental impacts 1.4 Recyclable and non-recyclable items 1.5 Different types of waste in RMG manufacturing 1.6 Hazardous waste in RMG production 1.7 Green habits to reduce waste
2. Underpinning Skills	2.1 Explaining principles of green practices in RMG manufacturing 2.2 Identifying sources of environmental impacts 2.3 Explaining re-cyclable and non-recyclable items 2.4 Minimizing resources use in the workplace 2.5 Identifying different types of waste in RMG manufacturing 2.6 Disposing hazardous waste in RMG production 2.7 Practicing green habits to reduce waste
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimizing resources use in the workplace 1.3 implementing waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: **UNDERSTAND QUALITY MANAGEMENT**	**Nominal Duration:** 20 Hrs**.**	**Unit Code:** SICIP-RMG-QCM-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to understand quality management. It specifically includes the task of interpreting quality, interpreting quality control and quality assurance and identifying roles and responsibilities of quality personnel.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Interpret quality	1.1 Quality is defined according to industry standards and organizational requirements. 1.2 **<u>Basic quality tools and equipment</u>** used for quality control and assurance is identified and described. 1.3 **<u>Quality dimensions</u>** are interpreted and explained. 1.4 The relationship between quality dimensions and the overall quality management system is understood. 1.5 Quality benchmarks and measurement criteria are identified with organizational goals for quality.
2. Interpret quality control and quality assurance	2.1 Importance of quality control systems is defined and explained. 2.2 **<u>Methods of quality control</u>** are identified and described. 2.3 **<u>Quality control procedures</u>** are identified and explained. 2.4 **<u>Quality assurance parameters</u>** are identified and described according to the specific needs of the product 2.5 The difference between quality control and quality assurance is explained.
3. Identify roles and responsibilities of quality personnel	3.1 The structure of the quality department is identified and described. 3.2 Reporting lines and communication flow within the quality department are identified. 3.3 **<u>Core responsibilities</u>** of quality personnel within the quality department are described.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Basic quality tools and equipment	1.1 Measuring tape 1.2 Button pull test machine 1.3 Needle detector machine 1.4 Pilling tester 1.5 Crock meter
2. Quality dimensions	2.1 Durability 2.2 Functionality 2.3 Aesthetic appeal 2.4 Performance 2.5 Reliability 2.6 Serviceability 2.7 Conformance
3. Methods of quality control	3.1 Testing 3.2 Inspection 3.3 Monitoring and auditing
4. Quality control procedures	4.1 Fabric and raw material inspection 4.2 Cutting quality control 4.3 In-process quality control (IPQC) 4.4 End-line quality inspection 4.5 Final quality control (FQC)
5. Quality assurance parameters	5.1 Dimension 5.2 Shrinkage and spirality 5.3 Strength 5.4 Colour fastness 5.5 Products safety 5.6 Customer satisfaction
6. Core responsibilities	6.1 Quality Inspections and Audits 6.2 Quality Standard Compliance 6.3 Defect identification 6.4 Process Improvement 6.5 Data Analysis 6.6 Documentation and Reporting

Curricular Content Guide

1. Underpinning Knowledge	1.1 Define quality 1.2 Basic quality tools and equipment 1.3 Quality dimensions 1.4 Overall quality management 1.5 Quality benchmarks and measurement criteria 1.6 Quality control procedures 1.7 Quality Control Parameters
2. Underpinning Skills	2.1. Interpreting quality 2.2. Interpreting quality dimensions 2.3. Identifying quality control procedures

	2.4. Interpreting quality control and quality assurance 2.5. Identifying roles and responsibilities of quality personnel
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Workplace procedures 4.3 Standard operating procedure 4.4 Workplace documents, signs and symbols 4.5 Projector 4.6 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted quality 1.2 interpreted quality dimensions 1.3 identified quality control procedures 1.4 interpreted quality control and quality assurance 1.5 Identified roles and responsibilities of quality personnel
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **IDENTIFY GARMENTS PROCESS OPERATION**	**Nominal Duration:** 30 Hrs**.**	**Unit Code:** SICIP-RMG-QCM-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to identify garments process operation. It specifically includes the task of identifying fabric and garment types, identifying sewing machine, identifying stitch and seam type, identifying types of trim and accessories and interpreting garment process operation.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Identify fabric and garment types	1.1 **<u>Fabric types</u>** are identified and classified based on their composition and texture. 1.2 **<u>Garment types</u>** are identified and categorized. 1.3 Garment parts and points are identified as per sample. 1.4 Fabric and garment classifications are compared and classified.
2. Identify sewing machine	2.1 **<u>Types of sewing machines</u>** are identified and classified based on their functions. 2.2 Key components of sewing machines are identified and described. 2.3 Machine features and specifications are explained.
3. Identify stitch and seam type	3.1 **<u>Stitch types</u>** are identified and categorized based on their function and application. 3.2 **<u>Seam types</u>** are identified and described according to their design and purpose. 3.3 The characteristics of each stitch type are explained.
4. Identify types of trim and accessories	4.1 **<u>Types of trim and accessories</u>** are identified and differentiated. 4.2 Use of trims and accessories are identified according to specification sheet.
5. Identify garments components	5.1. **<u>Garment components</u>** are identified based on standard operating processes. 5.2. Garment components are described.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Fabric types	1.1. Woven

	1.2. Non-woven 1.3. Knit 1.4. Braided
2. Garment types	2.1. Woven 2.1.1. Shirt 2.1.2. Pant 2.1.3. Jacket 2.1.4. Dress 2.1.5. Blazers 2.1.6. Waist coat 2.2. Knit 2.2.1. T-shirt 2.2.2. Pants/trouser 2.2.3. Polo-shirt 2.2.4. Hoodie jacket
3. Types of sewing machine	3.1. Chain stitch machine 3.1.1. Kansai multi-needle 3.1.2. Flat lock 3.1.3. Feed of the arm 3.1.4. Single needle chain stitch 3.2. Lock stitch machine 3.2.1. Single needle lock stitch 3.2.2. Double needle lock stitch 3.2.3. Overlock 3.2.4. Bar tuck 3.2.5. Button stitch 3.2.6. Button hole
4. Stitch type	4.1. Chain 4.2. Lock
5. Seam type	5.1. Super imposed seam 5.2. Lapped seam 5.3. Bound seam 5.4. Flat seam 5.5. Decorative seam 5.6. Edge neatening seam 5.7. Enclosed seam 5.8. Applied seam
6. Types of trim and accessories	6.1. Trims: 6.1.1. Sewing 6.1.2. Finishing 6.2. Accessories: 6.2.1. Informative 6.2.2. Decorative 6.2.3. Paper made 6.2.4. Plastic made
7. Garment components	7.1. Front part 7.2. Back part

	7.3. Assembling part

Curricular Content Guide

1. Underpinning Knowledge	1.1 Types of fabric and garment 1.2 Garment parts and points 1.3 Types of sewing machines 1.4 Stitch and seam type 1.5 Types of trim and accessories 1.6 Garments components
2. Underpinning Skills	2.1 Identifying fabric and garment types 2.2 Identifying sewing machine 2.3 Identifying stitch and seam type 2.4 Identifying types of trim and accessories 2.5 Identifying garments components
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Fabric and garment 4.4 Sewing machine 4.5 Trim and accessories 4.6 Projector 4.7 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified fabric and garment 1.2 Identified sewing machine 1.3 Identified stitch and seam 1.4 Identified types of trim and accessories 1.5 Identified garments components
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)

3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **IDENTIFY FAULTS AND DEFECTS**	Nominal Duration: 30 Hrs.	Unit Code: SICIP-RMG-QCM-03-O

Unit Descriptor:

This unit covers the skills, knowledge and attitudes required to identify faults and defects. It specifically includes the task of identifying fabric faults, identifying trim and accessories faults, identifying garment defects, classifying garment zones and defects and identifying product safety and integrity.

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify fabric faults	1.1 **Common fabric** faults are identified. 1.2 Fabric faults are categorized. 1.3 Solutions to repair and rectify fabric faults are identified
2. Identify trims and accessories faults	2.1 **Common trims and accessories faults** are identified. 2.2 Trim and accessories faults are classified. 2.3 Solutions to repair or replace trim and accessories faults are identified.
3. Identify garment defects	3.1 **Common garment defects** are identified. 3.2 Garment defects are categorized. 3.3 Solutions to repair or rectify garment defects are identified.
4. Classify garment zones and defects	4.1 **Garment zone classification** is identified and described. 4.2 **Garment defect classification** is carried out based on the specific garment zone. 4.3 Garment zone and defect classification records are maintained.
5. Identify product safety and integrity	5.1 **Product safety standards** are identified and described. 5.2 Potential safety risks in the product are recognized and evaluated. 5.3 Safety protocols and quality checks are followed throughout the production process.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Common fabric faults	1.1 Oily stains 1.2 Dust to surface 1.3 Improper cleaning 1.4 Very high yarn twist 1.5 Needle broken

	1.6 Uneven yarn
2. Common trims and accessories faults	2.1 Unmatched colour of thread 2.2 Broken button and zipper 2.3 Short zippers 2.4 Wrong labels 2.5 Improper size labels 2.6 Printing mistake of labels and cartons 2.7 Broken polybag 2.8 Wrong hanger 2.9 Improper embroideries and prints
3. Common garment defects	3.1 Seam puckering 3.2 Spirality 3.3 Broken buttons 3.4 Broken stitching 3.5 Defective snaps 3.6 Different shades (within the same garment) 3.7 Dropped stitch 3.8 Skipped stitch 3.9 Exposed stitch 3.10 Exposed raw edges 3.11 Fabric defects 3.12 Holes 3.13 Inoperative zipper 3.14 Loose sewing threads 3.15 Misaligned buttons and holes 3.16 Needle cuts 3.17 Open seams 3.18 Loose thread 3.19 Stain/spot 3.20 Unfinished buttonhole 3.21 Zipper too short
4. Garment zone classification	4.1 Zone A 4.2 Zone B 4.3 Zone C/Zone D
5. Garment defect classification	5.1. Critical defects 5.2. Major defects 5.3. Minor defects
6. Product safety standards	6.1 Compliance with Regulations 6.2 Material Safety 6.3 Durability and Strength 6.4 Inappropriate labelling and misinformation 6.5 Product recall

Curricular Content Guide

1. Underpinning Knowledge	1.1 Fabric faults 1.2 Trims and accessories

	1.3 Garment defects 1.4 Garment zones and defects 1.5 Product safety and integrity 1.6 Safety protocols and quality checks
2. Underpinning Skills	2.1. Identifying fabric faults 2.2. Identifying trim and accessories faults 2.3. Identifying garment defects 2.4. Classifying garment zones and defects 2.5. Identifying product safety and integrity.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Fabric and garment 4.4 Sewing machine 4.5 Trim and accessories 4.6 Projector 4.7 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified fabric faults 1.2 Identified trims and accessories faults 1.3 Identified garment defects 1.4 Classified garment zones and defects 1.5 Identified product safety and integrity
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **INSPECT FABRIC, TRIMS AND ACCESSORIES**	**Nominal Duration:** 20 Hrs**.**	**Unit Code:** SICIP-RMG-QCM-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to Inspect fabric, trims and accessories. It specifically includes the task of carrying out fabric inspection and carrying out trim and accessories inspection		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Carry out fabric inspection	1.1 **Fabric inspection methods** are identified and described. 1.2 **Inspection tools and equipment** are selected and used. 1.3 Fabric inspection is performed according to buyer guidelines. 1.4 Inspection reports are prepared for internal and buyer review.
2. Carry out trims and accessories inspection	2.1 **Trims and accessories inspection methods** are identified and described. 2.2 Trims and accessories inspection is performed in accordance with the buyer's guidelines. 2.3 Inspection reports are prepared.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Fabric inspection methods	1.1 4-point system 1.2 10-point system 1.3 Dallas system
2. Inspection tools and equipment	2.1 Measuring tools 2.2 Fabric inspection machine 2.3 Light box
3. Trim and accessories inspection methods	3.1 Countable item inspection method 3.2 Uncountable item inspection method

Curricular Content Guide

1. Underpinning Knowledge	1.1 Fabric inspection methods 1.2 Inspection tools and equipment 1.3 Performing fabric inspection 1.4 Inspection reports 1.5 Trim and accessories inspection methods
2. Underpinning Skills	2.1. Identified fabric inspection methods

	2.2. Identified trims and accessories inspection methods 2.3. Performing trims and accessories inspection 2.4. Preparing inspection reports
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Required tools, Equipment and facilities 4.4 Relevant Materials for proposed activity 4.5 Manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified fabric inspection methods 1.2 identified trims and accessories inspection methods 1.3 performed trims and accessories inspection 1.4 prepared inspection reports
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT SECTION WISE QUALITY CONTROL	Nominal Duration: 60 Hrs.	Unit Code: SICIP-RMG-QCM-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out section wise quality control. It specifically includes the task of performing quality control in sample, performing quality control in washing, performing quality control in cutting, performing quality control in embellishment, performing quality control in sewing and performing quality control in finishing.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Perform quality control in sample	1.1 **Types of samples** are identified. 1.2 **Technical issues** are identified. 1.3 Quality control procedures are followed to ensure that the sample meets all the **required criteria.**
2. Perform quality control in washing	2.1 Washing quality standards are identified and interpreted. 2.2 Washing methods are identified. 2.3 Inspection of washed garments is carried out. 2.4 Washing defects are identified and recorded. 2.5 Washing quality control reports are prepared.
3. Perform quality control in cutting	3.1 **Pattern and marker checking** procedure is explained. 3.2 Inspection of pattern and marker is carried out. 3.3 Cut fabric checking procedure is identified and described. 3.4 Inspection of cut fabric is carried out. 3.5 Cutting quality reporting formats are identified.
4. Perform quality control in embellishment	4.1 Embellishment quality standards are identified and interpreted. 4.2 Embellishments are inspected for defects. 4.3 Defective embellishments are identified, recorded, and categorized. 4.4 Quality control documentation for embellishments is prepared.
5. Perform quality control in sewing	5.1 Inline inspection procedure is explained. 5.2 Endline inspection procedure is explained. 5.3 Inspection of inline and endline is carried out. 5.4 Sewing quality control reports are prepared.
6. Perform quality control in finishing	6.1. Finishing quality standards are identified and interpreted. 6.2. **Finishing procedures** are followed and carried out. 6.3. Quality control reports for finishing are prepared.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Types of samples	1.1 Development/styling 1.2 Fit 1.3 Size set 1.4 Pre-production 1.5 Wash test 1.6 Production 1.7 Shipment
2. Technical issues	2.1 Shrinkage 2.2 Master patter 2.3 Pattern grading 2.4 Related issues
3. Required criteria	3.1 Match with specification 3.2 Fabrication 3.3 Trims and accessories 3.4 Workmanship 3.5 Measurement 3.6 Getup/presentation of the sample 3.7 Sample information
4. Pattern and marker checking	4.1 Number of pattern and garment pieces 4.2 Number of garments 4.3 Parts alignment 4.4 Fabric consumption 4.5 Marker efficiency
5. Finishing procedure	5.1 Ironing 5.2 Folding 5.3 Packing and carton

Curricular Content Guide

1. Underpinning Knowledge	1.1 Types of samples 1.2 Technical issues 1.3 Quality control procedures 1.4 Washing methods 1.5 Inspection of washed garments 1.6 Washing defects 1.7 Inspection of pattern and marker 1.8 Quality control documentation 1.9 Inline inspection procedure 1.10 Finishing procedures 1.11 Quality control reports
2. Underpinning Skills	2.1 Performing quality control in sample 2.2 Performing quality control in washing 2.3 Performing quality control in cutting 2.4 Performing quality control in embellishment 2.5 Performing quality control in sewing

	2.6 Performing quality control in finishing
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Required tools, Equipment and facilities 4.4 Relevant Materials for proposed activity 4.5 Manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 performed quality control in sample 1.2 performed quality control in washing 1.3 performed quality control in cutting 1.4 performed quality control in embellishment 1.5 performed quality control in sewing 1.6 performed quality control in finishing
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM FINAL INSPECTION AND AUDIT**	Nominal Duration: 40 Hrs.	Unit Code: SICIP-RMG-QCM-06-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform final inspection and audit. It specifically includes the task of preparing for final inspection, conducting final product inspection and interpret technical audit system.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for final inspection	1.1. Acceptable Quality Level (AQL) is identified and defined. 1.2. **AQL chart** is identified as per standard operating procedure 1.3. Importance and use of AQL chart are described.
2. Conduct final product inspection	2.1 Procedure for packing list verification and carton selection is identified. 2.2 Garment selection is carried out as per AQL and standard operating procedure. 2.3 **Inspection methodology** to be utilised is identified and explained. 2.4 **Final inspection** is carried out as per standard operating procedure.
3. Interpret technical audit system	3.1. **Technical audit system** is understood and interpreted. 3.2. Key components of the technical audit system are identified. 3.3. Audit objectives and their relevance to the organization's goals are explained. 3.4. Audit process steps are described. 3.5. Technical criteria and benchmarks are identified.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. AQL chart	1.1. Lot or batch size 1.2. Sample size code letter 1.3. Sample size 1.4. Acceptable quality level (AQL)
2. Inspection methodology	2.1. Environment checking of inspection room 2.2. Packing list and carton verification

	2.3. Sample lot selection
3. Final inspection	3.1 Quantity of the order 3.2 Matching with final sample 3.3 Garments checking as per AQL 3.4 Needle detection 3.5 Measurement check 3.6 Inspection report
4. Technical audit system	4.1. Good Manufacturing Practice (GMP) 4.2. Factory Capacity and Capability Audit (FCCA) 4.3. Process validation (PV) 4.4. Supplier Excellence (SE) 4.5. ISO 9001:2015 (QMS)

Curricular Content Guide

1. Underpinning Knowledge	1.1 Acceptable Quality Level (AQL) 1.2 AQL chart 1.3 Inspection methodology 1.4 Final inspection 1.5 Technical audit system 1.6 Audit process steps
2. Underpinning Skills	2.1 Identifying Acceptable Quality Level (AQL) 2.2 Identifying AQL chart 2.3 Preparing for final inspection 2.4 Conducting final product inspection 2.5 Interpret technical audit system 2.6 Carrying out Final inspection
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Required tools, Equipment and facilities 4.4 Relevant Materials for proposed activity 4.5 Manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified Acceptable Quality Level (AQL) 1.2 identified AQL chart 1.3 prepared for final inspection 1.4 conducted final product inspection 1.5 interpreted technical audit system 1.6 carried out final inspection
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **INTERPRET TQM AND LEAN MANAGEMENT**	**Nominal Duration:** 50 Hrs**.**	**Unit Code:** SICIP-RMG-QCM-07-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to interpret TQM and lean management. It specifically includes the task of defining TQM and lean manufacturing, identifying TQM tools, identifying lean manufacturing waste, identifying lean tools and interpret six sigma.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Define TQM and lean manufacturing	1.1. Concept of Total Quality Management (TQM) is defined and explained. 1.2. Principles and objectives of TQM are described. 1.3. Concept of Lean Manufacturing is defined and interpreted. 1.4. Principles of lean manufacturing is explained. 1.5. The relationship between TQM and Lean Manufacturing is described.
2. Identify TQM tools	2.1 **Total Quality Management (TQM) tools** are identified and listed according to industry standards. 2.2 Purpose and function of each TQM tool are described. 2.3 Selection criteria for appropriate TQM tools based on specific quality issues are explained. 2.4 Use of TQM tools to analyze and solve quality problems is understood. 2.5 Importance of TQM tools in supporting continuous improvement and decision-making is recognized.
3. Identify lean manufacturing waste	3.1. Types of lean manufacturing waste are identified and described. 3.2. Impact of each waste type on production efficiency, cost, and quality is explained. 3.3. Techniques for detecting and measuring waste within lean manufacturing systems are identified. 3.4. Relevant organizational policies and industry standards related to waste management are identified and explained.
4. Identify lean tools	4.1. Common lean tools are identified and listed according to industry standards. 4.2. Purpose and application of each lean tool are described in relation to waste reduction and process improvement. 4.3. Selection of appropriate lean tools is explained based on specific process needs and organizational goals.

5. Interpret six sigma	5.1. Six Sigma concepts and principles are identified and interpreted. 5.2. DMAIC framework is identified and interpreted. 5.3. Benefits of implementing Six Sigma in manufacturing processes are explained. 5.4. The key tools and techniques used within the DMAIC framework are described. 5.5. Application of Six Sigma is interpreted.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Total Quality Management (TQM) tools	1.1 7 quality tools: 1.2 Pareto charts 1.3 Cause-and-effect diagrams (fishbone diagrams) 1.4 Control charts 1.5 Histograms 1.6 Scatter diagrams 1.7 Flowcharts 1.8 Check sheets 1.9 Plan Do Check Act (PDCA) cycle 1.10 Failure Mood and Effects Analysis (FMEA)
2. Types of lean manufacturing waste	2.1 Defects 2.2 Overproduction 2.3 Waiting 2.4 Non-utilized talent 2.5 Transportation 2.6 Inventory 2.7 Motion 2.8 Extra-processing
3. Common lean tools	3.1 5S 3.2 Seiri (Sort) 3.3 Seiton (Set in order) 3.4 Seiso (Shine) 3.5 Seiketsu (Standardize) 3.6 Shitsuke (Sustain) 3.7 Kaizen (Continuous improvement) 3.8 Value Stream Mapping (VSM) 3.9 Kanban (Flexible) 3.10 Poka-Yoke (Mistake-proofing) 3.11 Just-In-Time (JIT) 3.12 3M model 3.13 Muda (Waste) 3.14 Mura (Unevenness) 3.15 Muri (Overburden)
4. DMAIC framework	4.1 Define 4.2 Measure

	4.3 Analyze 4.4 Improve 4.5 Control
5. Benefits of implementing Six Sigma	5.1 Less variation 5.2 Minimizing of waste 5.3 Process Improvement 5.4 Ensure quality 5.5 Minimizing defects 5.6 Increase customer satisfaction

Curricular Content Guide

1. Underpinning Knowledge	1.1 Define Total Quality Management (TQM) 1.2 Principles and objectives of TQM 1.3 Concept of Lean Manufacturing 1.4 Total Quality Management (TQM) tools 1.5 T Common lean tools ypes of lean manufacturing waste 1.6 Six Sigma concepts and principles 1.7 DMAIC framework 1.8 Application of Six Sigma
2. Underpinning Skills	2.1 Defining TQM and lean manufacturing 2.2 Identifying TQM tools 2.3 Identifying lean manufacturing waste 2.4 Identifying lean tools 2.5 Interpreting six sigma
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Required tools, Equipment and facilities 4.4 Relevant Materials for proposed activity 4.5 Manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Defined TQM and lean manufacturing 1.2 Identified TQM tools 1.3 Identified lean manufacturing waste 1.4 Identified lean tools

	1.5 Interpreting six sigma
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **APPLY ROOT CAUSE ANALYSIS TECHNIQUES**	**Nominal Duration:** 50 Hrs**.**	**Unit Code:** SICIP-RMG-QCM-08-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply root cause analysis techniques. It specifically includes the task of analysing data and identifying root causes, implement root cause analysis techniques, implementing corrective and preventive actions and preparing report analysis results.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Analyse Data and Identify Root Causes	1.1. Relevant data are collected and organized systematically to support analysis. 1.2. Appropriate **<u>data analysis methods</u>** are applied. 1.3. Findings from data analysis are documented. 1.4. Root causes of identified issues are determined.
2. Implement Root Cause Analysis Techniques	2.1. **<u>Root cause analysis techniques</u>** are identified and selected based on specific quality control issue. 2.2. Data requirements for each root cause analysis technique are determined to ensure effective application. 2.3. Selected root cause analysis techniques are applied systematically to analyze problems.
3. Implement Corrective and Preventive Actions	3.1 Corrective actions are identified to address the root causes of quality issues. 3.2 Preventive actions are proposed. 3.3 Corrective and preventive action plans are prepared. 3.4 Effectiveness of corrective actions is monitored.
4. Prepare Report Analysis Results	4.1 Analysis results are reviewed and verified. 4.2 Report format is selected and followed according to organizational standards. 4.3 Final report is prepared and submitted to relevant stakeholders.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Data analysis methods	1.1. Check sheet 1.2. Top 3 defects analysis 1.3. Traffic Light System (TLS)
2. Root cause analysis techniques	2.1. 5 why 2.2. Fishbone diagram 2.3. Pareto chart

Curricular Content Guide

1. Underpinning Knowledge	1.1 Data analysis methods 1.2 Root cause analysis techniques 1.3 Corrective actions 1.4 Preventive actions
2. Underpinning Skills	2.1 Analyzing data 2.2 Identifying root causes 2.3 Implementing root cause analysis techniques 2.4 Implementing corrective and preventive actions
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Required tools, Equipment and facilities 4.4 Relevant Materials for proposed activity 4.5 Manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 analyzed data 1.2 identified root causes 1.3 implemented root cause analysis techniques 1.4 implemented corrective and preventive actions
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	**Quality Control Management**
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sqm) OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

S.N.	Major Equipment and Training facilities	Required facilities
1.	Computer/Laptop	2
2.	Multimedia Projector	1
3.	Internet connectivity	1
4.	Display Board Samples of trims (sewing trims, finishing trims) and accessories (informative, decorative, paper, plastic)	10
5.	Fabric defect samples (yarn defects, weaving/knit defects, dyeing/printing/finishing defects)	20
6.	Samples of different embellishments defect (prints, embroidery)	20
7.	Samples of garment defects	20
8.	Garments Inspection Table with Lighting System	2
9.	Fabric Inspection Table with Lighting System (min. 6 feet)	2
10.	Fabric Inspection Machine	1
11.	Tensile testers for fabric strength testing	1
12.	Pilling testers for fabric surface quality evaluation	1
13.	Abrasion testers for wear and tear simulation	1
14.	Colour fastness testers for colour durability assessment	1
15.	Gray scale	6
16.	Measurement tape	26
17.	Light Box	1
18.	GSM Cutter	2
19.	Button Pull Test Machine	1

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on **Quality Control Management** the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.